Geometry
Unit 7
Lesson H2 Homework

Name Answer Key
Date____________________
Period____________________

In $\triangle TBN$, $m\angle TNB = 46°$, $TN = 17.2$, and $BN = 19.4$. Determine the measurement of each part of $\triangle TBN$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$\angle NTB$	$75.2°$
2.	$\angle TBN$	$59°$
3.	$\overline{BT}$	14.4

1) $19.4^2 = 14.4^2 + 17.2^2 - 2(14.4)(17.2)\cos T$
$376.36 = 207.36 + 295.84 - 495.36\cos T$
$-126.84 = -495.36\cos T$
$0.256 = \cos T$
$T = 75.2°$

2) $17.2^2 = 14.4^2 + 19.4^2 - 2(14.4)(19.4)\cos B$
$295.84 = 207.36 + 376.36 - 558.72\cos B$
$-287.88 = -558.72\cos B$
$0.515 = \cos B$
$B = 59°$

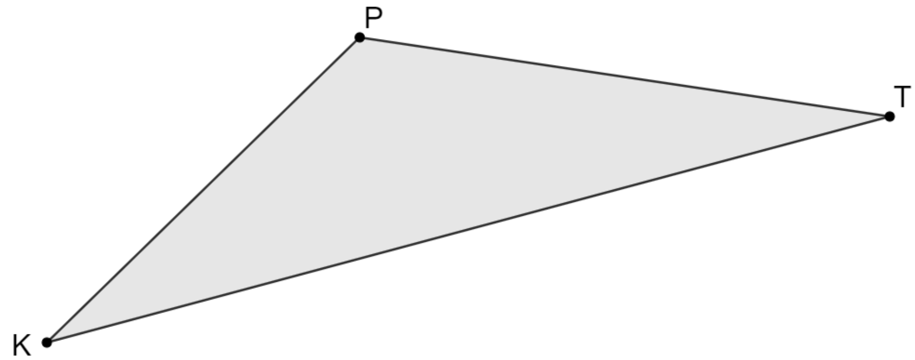
3) $BT^2 = 19.4^2 + 17.2^2 - 2(19.4)(17.2)\cos 46 = 208.61$
$BT = 14.4$

In $\triangle KRM$, $KR = 5$, $MK = 5$, and $MR = 7.826$. Determine the measurement of each part of $\triangle KRM$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$\angle RKM$	$103°$
5.	$\angle MRK$	$38.5°$
6.	$\angle KMR$	$38.5°$

$5^2 = 7.826^2 + 5^1 - 2(7.826)(5)(\cos R)$
$25 = 61.246 + 25 - 78.26\cos R$
$-61.246 = -78.26\cos R$
$0.783 = \cos R$
$\angle R = 38.5°, \quad \angle M = 38.5°$
$\angle K = 180 - 2(38.5) = 103°$

When Camila visits the zoo she walks from the kangaroos, point K, to the Pandas, point P, then to the tigers, point T, before returning to the Kangaroos. The path from the kangaroos to the pandas is 398 yards long. The path from the pandas to the tigers is 488 yards long. The path from the tigers to the kangaroos is 795 yards long. Determine each angle measure.



	Angle	Measurement
7.	$\angle KPT$	$127.3°$
8.	$\angle PTK$	$23.5°$
9.	$\angle TKP$	$29.2°$

10. In $\triangle CYL$, $CY = 193$, $LY = 95$, and $CL = 168$. Complete the statement about $\triangle CYL$.

$\triangle CYL$ is a

- o right
- o acute
- o obtuse
- o right

triangle because

- o one of the angles is
- o all of the angles are

- o equal to $90°$.
- o less than $90°$.
- o greater than $90°$.